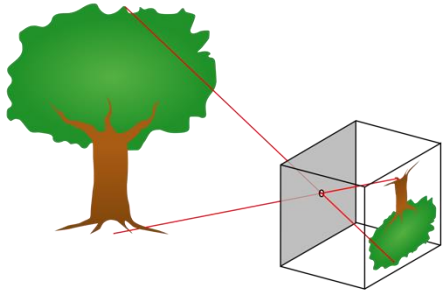
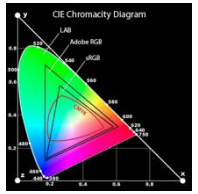
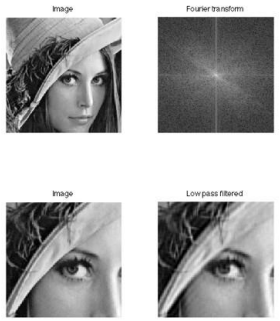


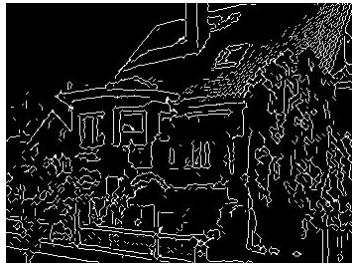
wikipedia.org





Computer Vision

Image Formation and Image Processing Basics



© wikipedia

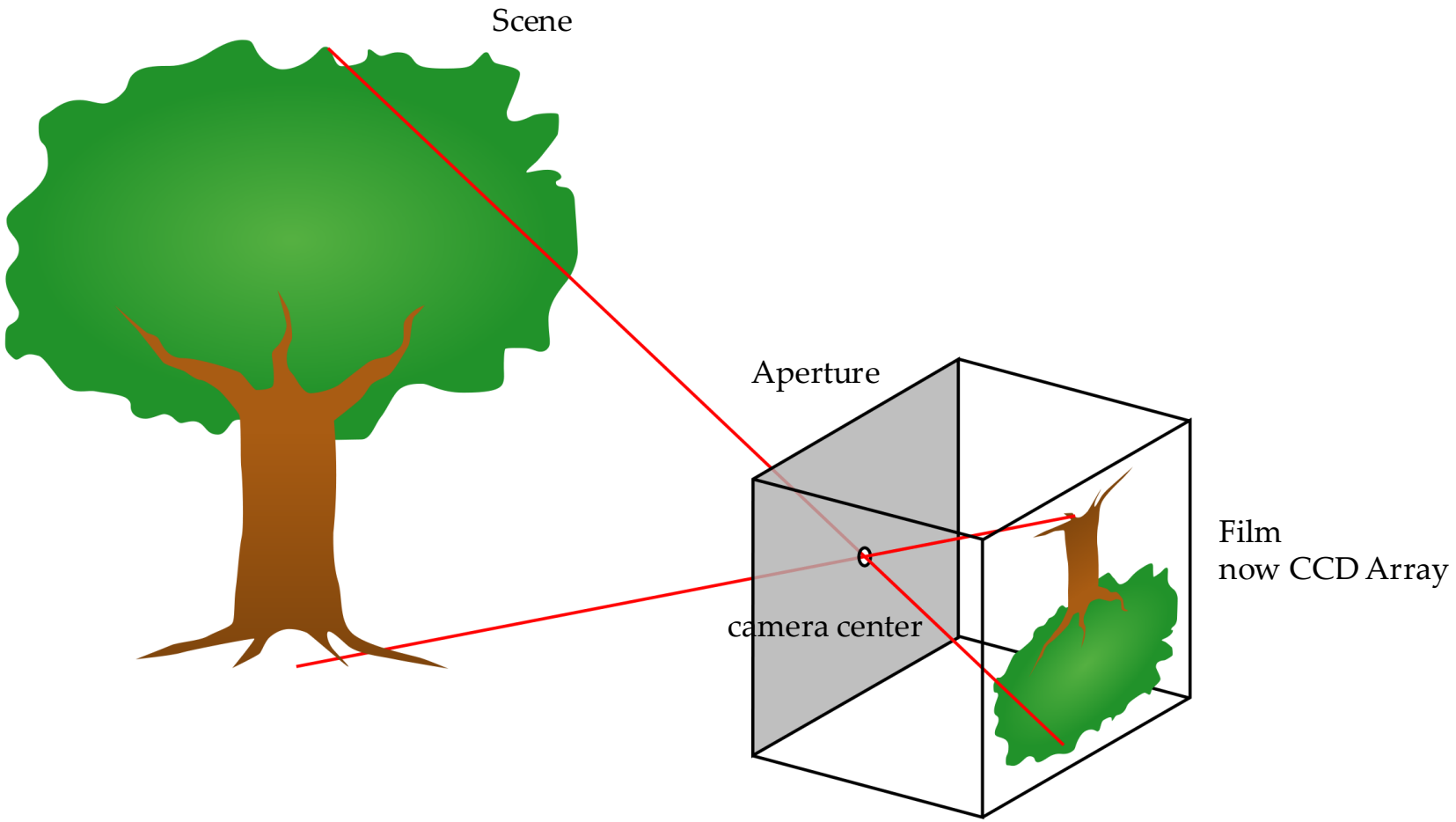
Master Medieninformatik / Technische Informatik
Prof. Dr. Kristian Hildebrand
khildebrand@bht-berlin.de

Goals for today

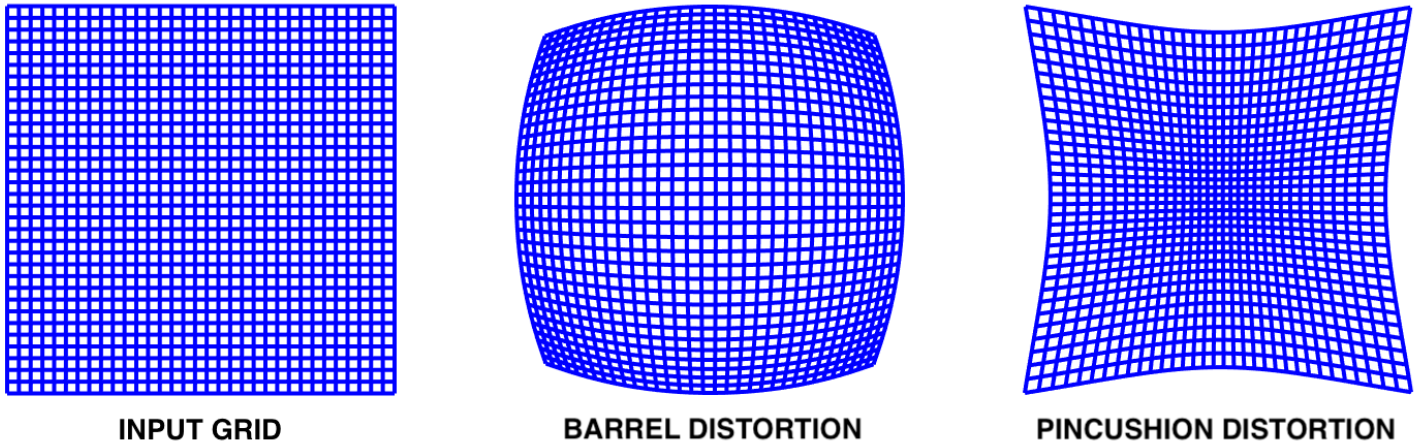
- Image formation (Extrinsic and Intrinsic Camera Parameters)
- Image representation
- Image processing
  - Filter
  - Algorithms, e.g. Canny-Edge, Color Quantization

- ***Assignment 1 is out***

Image formation

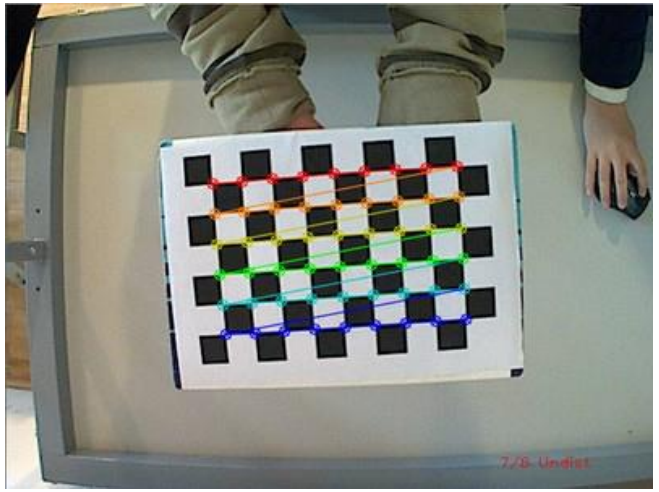


wikipedia.org

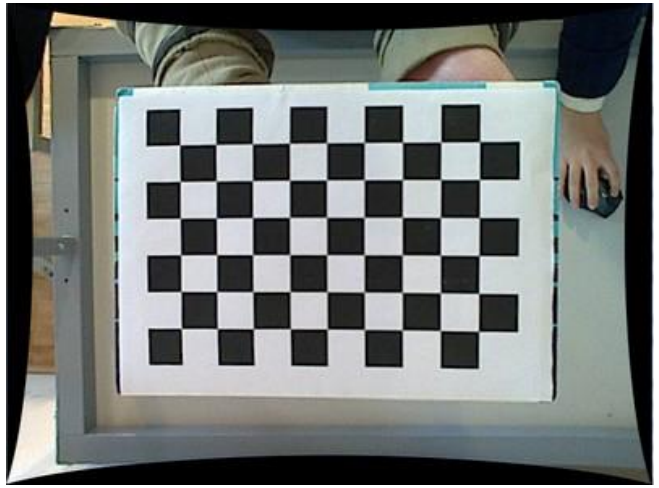


$$x_{distorted} = x(1 + k_1 r^2 + k_2 r^4 + k_3 r^6)$$
$$y_{distorted} = y(1 + k_1 r^2 + k_2 r^4 + k_3 r^6)$$

https://docs.opencv.org/3.4/d4/d94/tutorial_camera_calibration.html



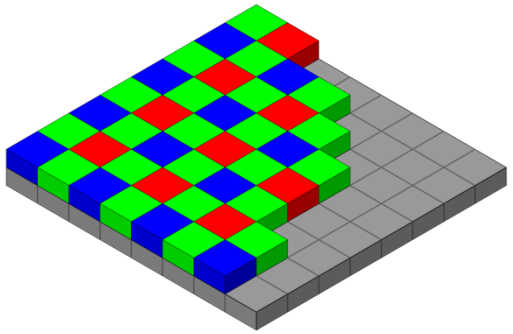
Undistort
e.g. with OpenCV

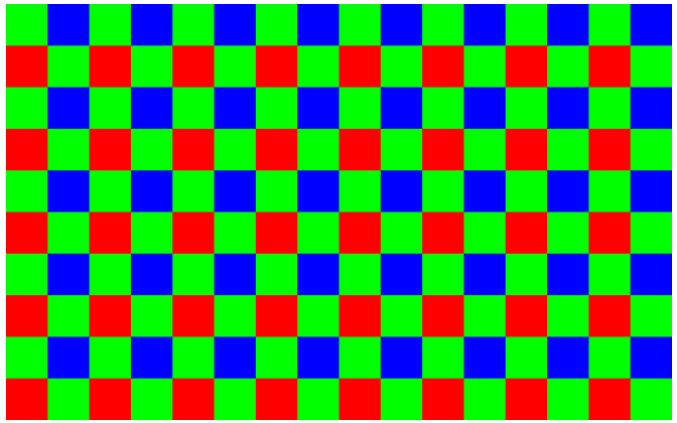


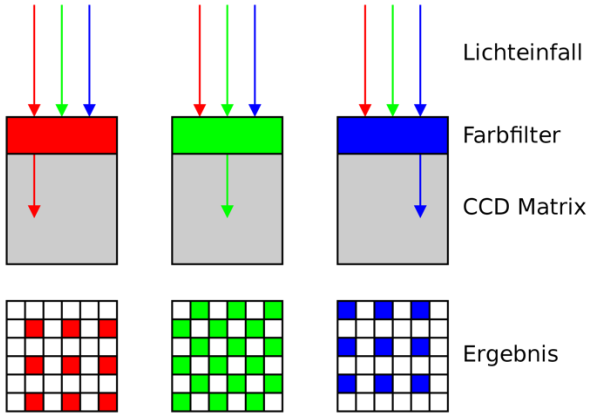
https://www.youtube.com/watch?v=ViPN810E0SU&ab_channel=OpenCVTutorials

Intrinsic and Extrinsic Camera Parameters

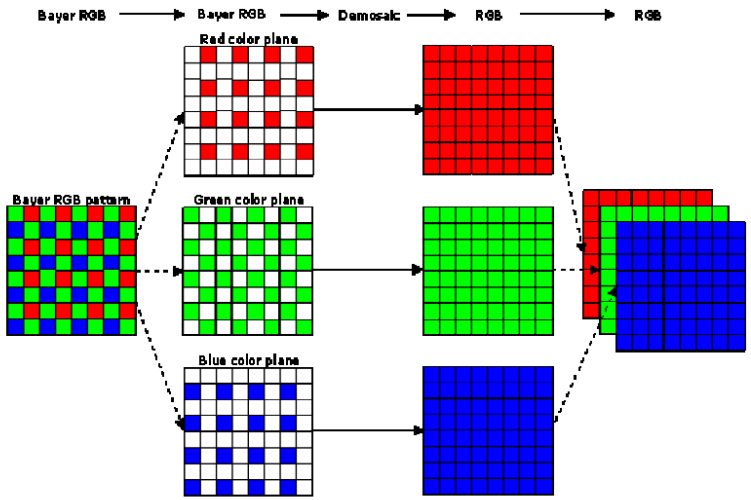
$$\begin{pmatrix} u \\ v \\ 1 \end{pmatrix} = K(R \cdot \begin{pmatrix} X \\ Y \\ Z \end{pmatrix} + t)$$







https://de.wikipedia.org/wiki/Bayer-Sensor



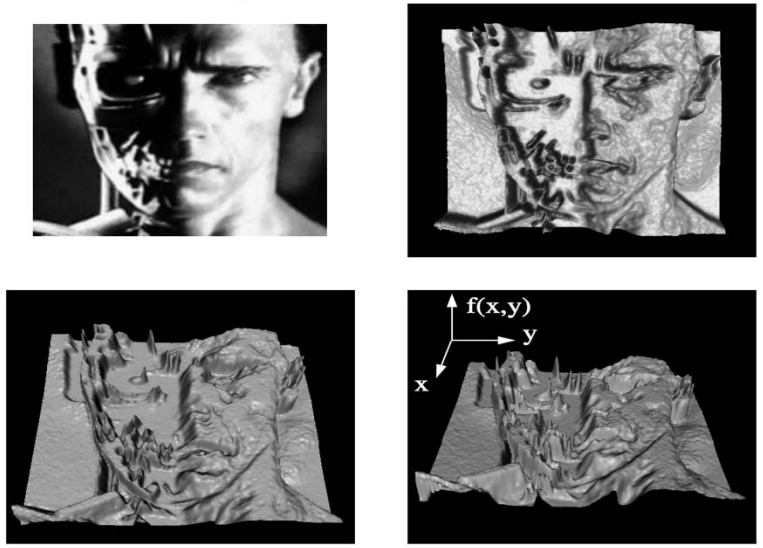
http://dias.library.tuc.gr/view/manf/25161

Image representation

Images are functions

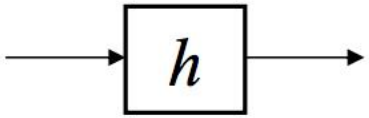
- **convex, continuous, differentiable**
- *I(x,y)* are intensities at location *(x,y)*

- change of signal through transformation function *h*



N. Snavely

- $g(x) = h(f(x))$



Invert, Histogram, Contrast
etc.

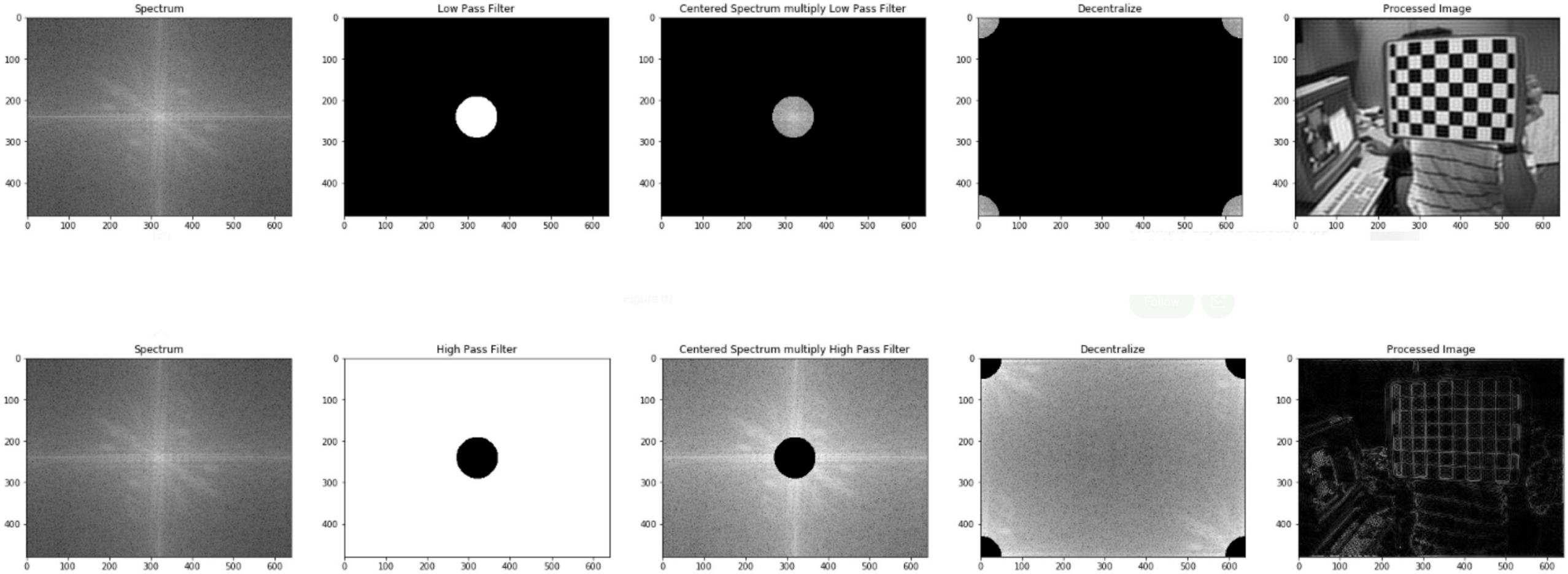
Fredo Durand, MIT

Fourier Transform

$$F(k,l) = \sum_{i=0}^{N-1} \sum_{j=0}^{N-1} f(i,j) e^{-i2\pi(\frac{ki}{N} + \frac{lj}{N})}$$

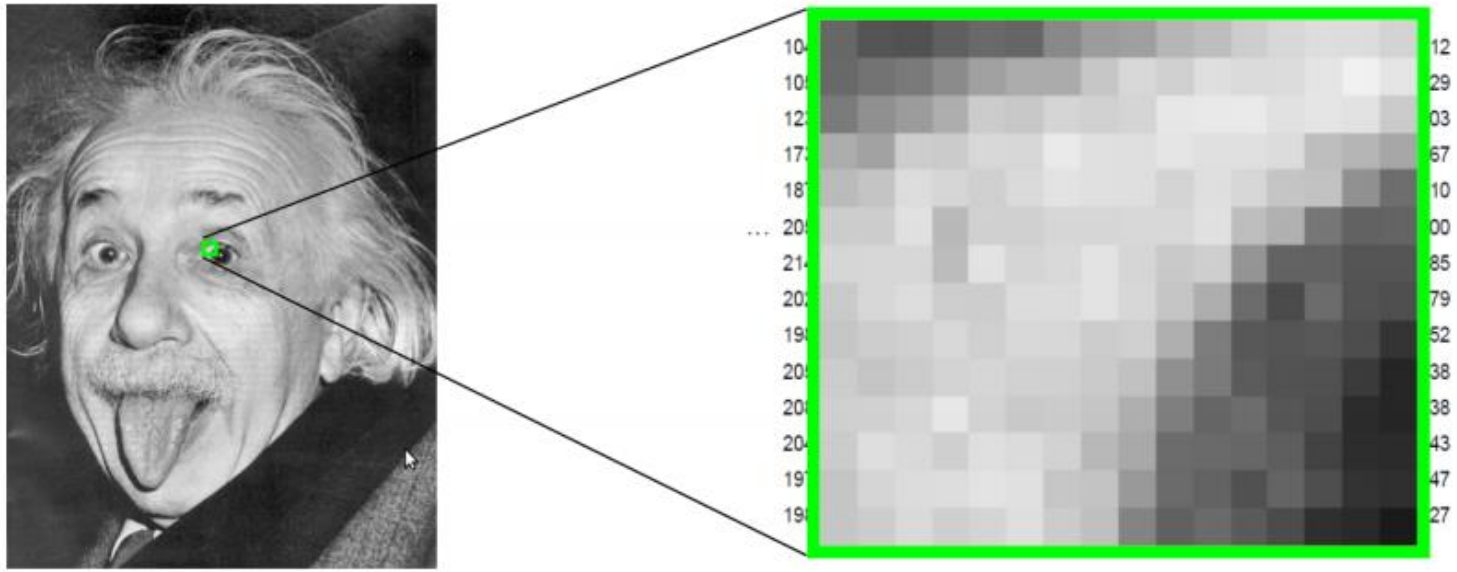
$$e^{ix} = \cos x + i \sin x$$

http://www.jezzamon.com/fourier/index.html



https://hicraigchen.medium.com/digital-image-processing-using-fourier-transform-in-python-bcb49424fd82

Image representation



D. Schlesinger, TU Dresden

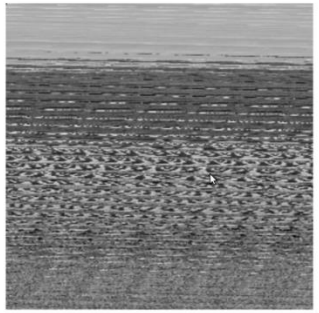
Matrices

or vectors

$$A_{n\times m} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & \boxed{a_{nm}} \end{bmatrix}$$

Pixel's intensity value

Images are Matrices

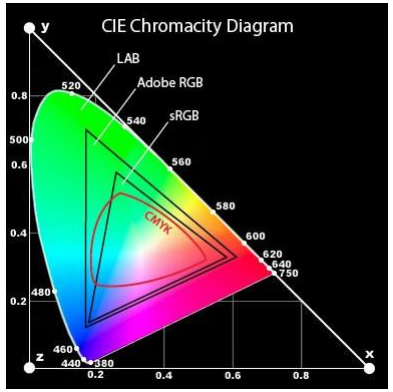


D. Schlesinger, TU Dresden

- similar vectors are not necessarily similar images
- similar images are not necessarily similar vectors
- Representation, geometric or color transformation (RGB vs. LAB color space)

RGB – Color space

LAB – Color space



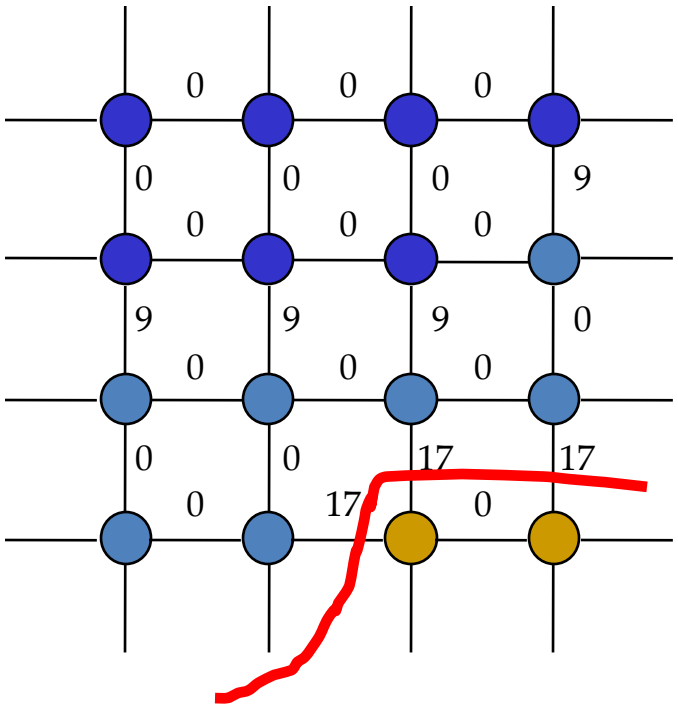
https://www.photo.net/learn/using-lab-color-adjustments/

Youtube Video in Moodle that explains color spaces

Compute distances between colors in LAB (diffs correspond more naturally to human color perception)

Images as graphs

- **Pixel are nodes with 4- (or 8-) neighbors**
- **Edges are weights**
  - e.g. color distances between pixels
- **Graph is a grid**
  - nodes can be labeled for segmentation etc.
- **Algorithmen:**
  - Graph Cut
  - Conditional Random Fields (CRF)



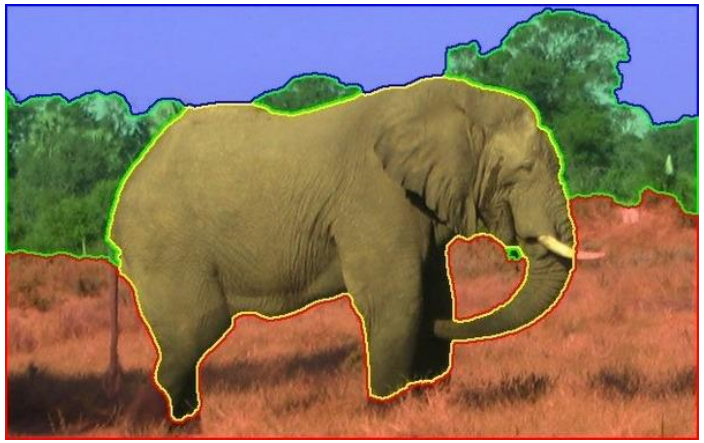
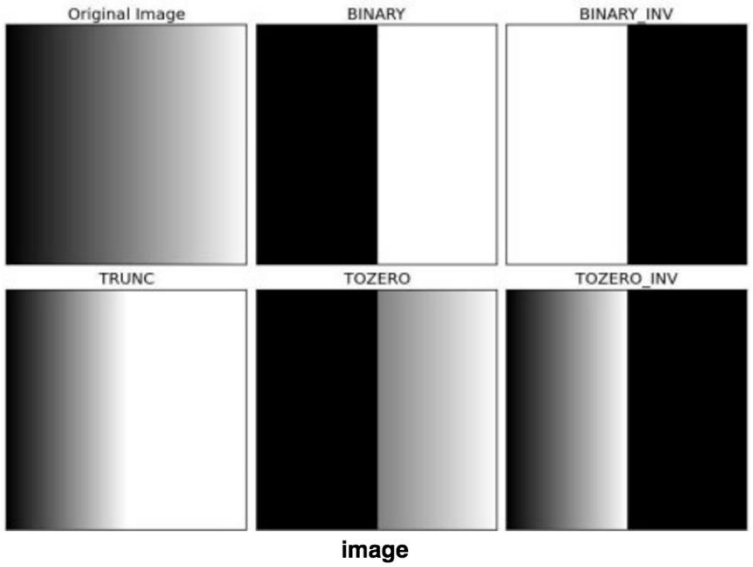
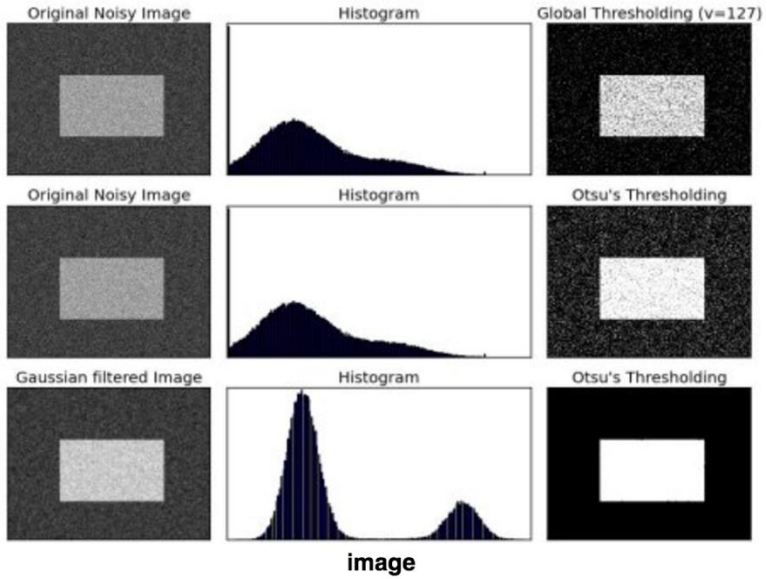


Image Processing
What is OpenCV?

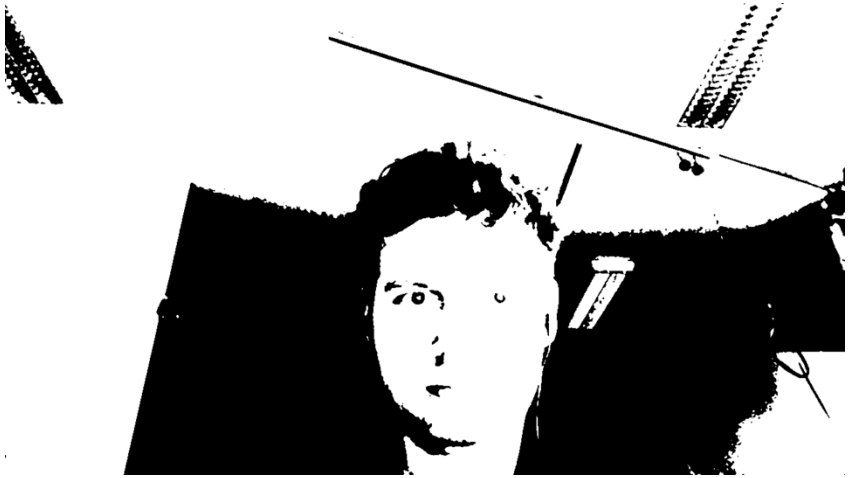
Image processing

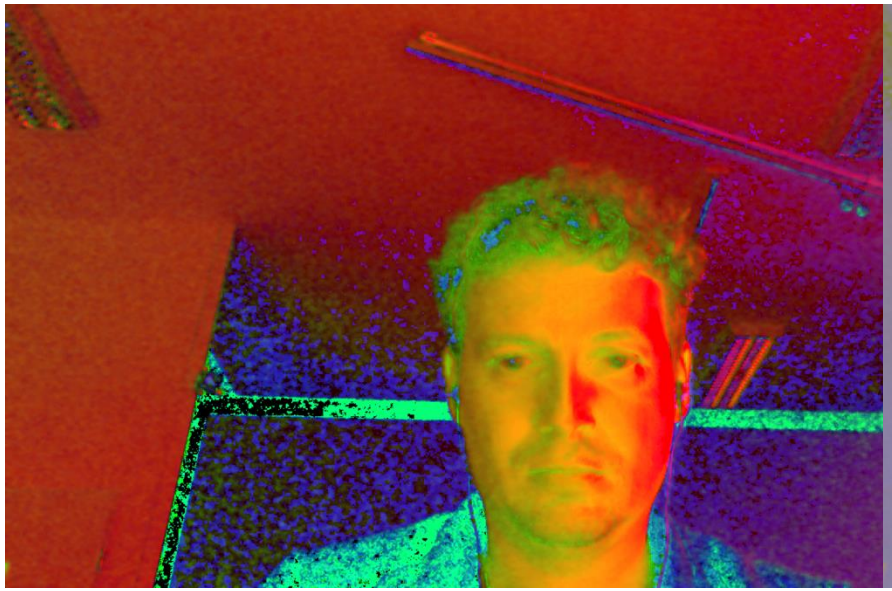
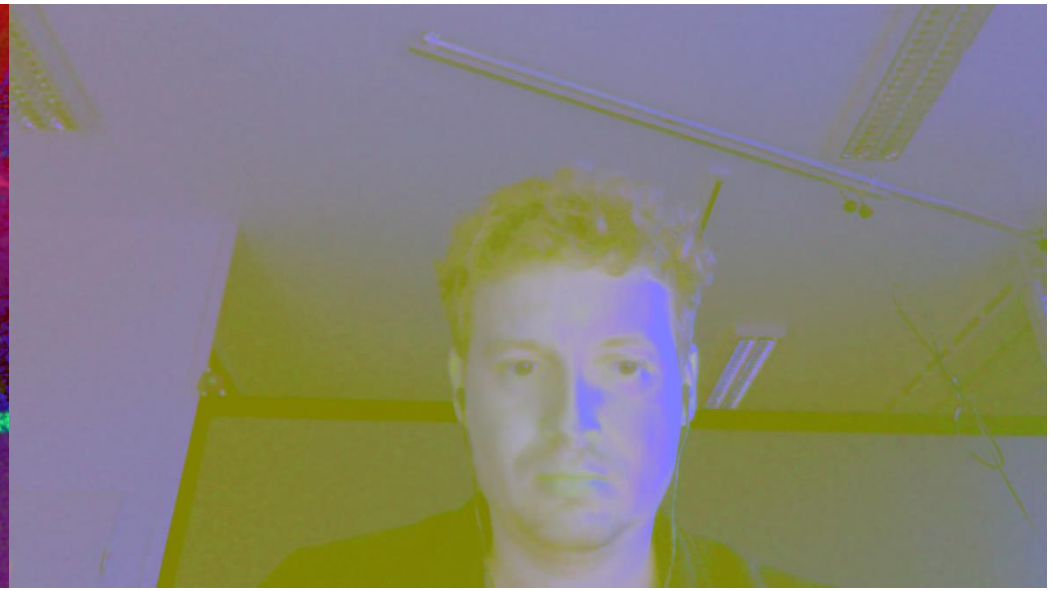
- Thresholding
- Changing Colorspaces
- Convolution + Image Gradients
- Canny Edge Detection
- Color Quantization with k-means



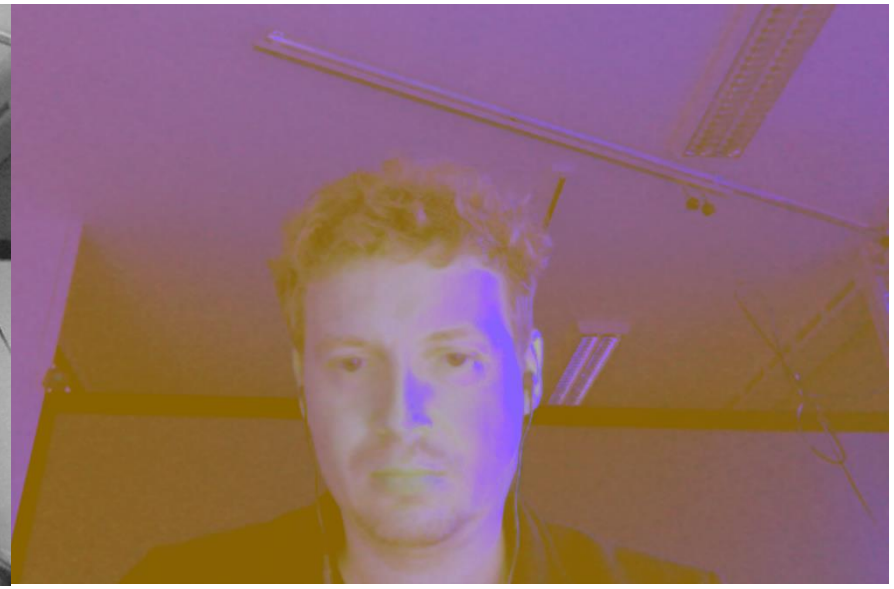


Thresholding



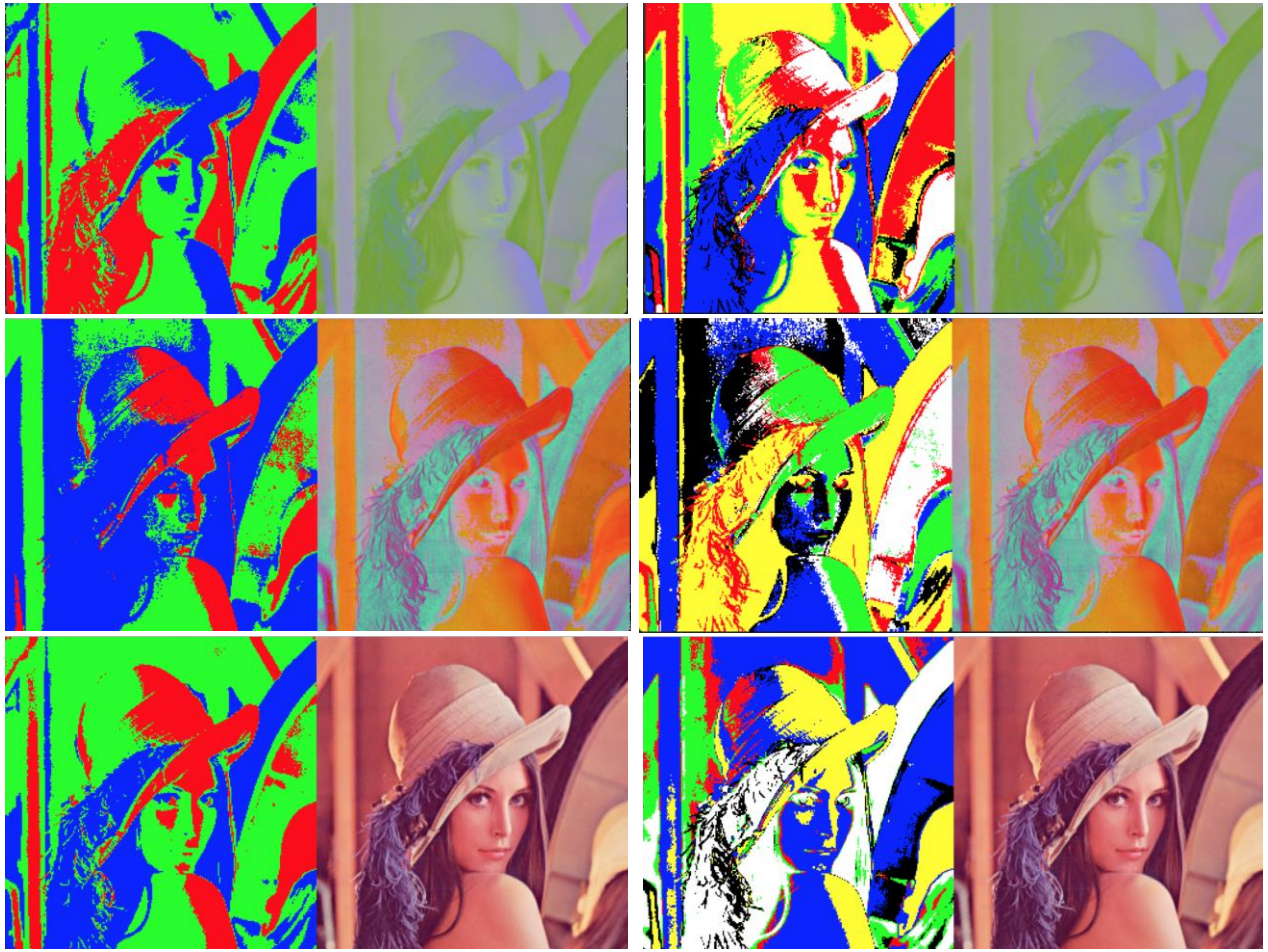



Colorspaces



Color Quantization

Original k=4 k=16 k=32 k=64

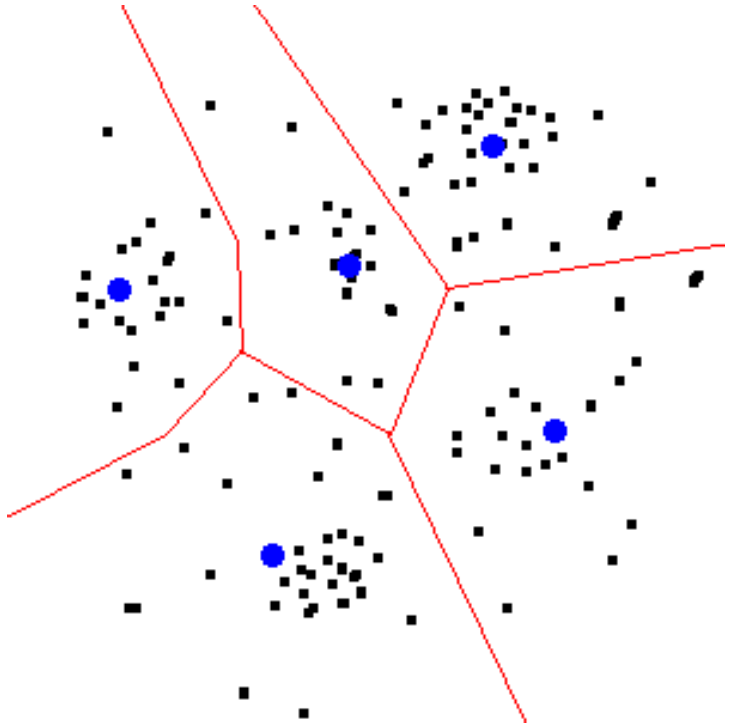


(a) (b)

k-means

- important **clustering method** in CS
- partitions N **data points** in k **clusters** $S = \{S_1, S_2, \ldots S_k\}$
- each **data point belongs to cluster with nearest mean**
  - mean of the cluster is cluster center/ representative

$$\arg\min_{s} \sum_{i=1}^{k} \sum_{x \in s_i} \|x - \mu_i\|^2$$



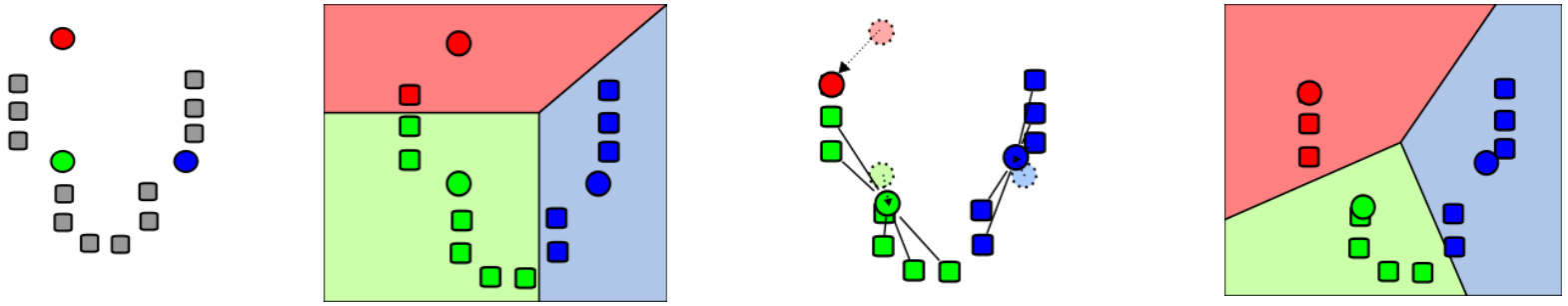
http://mnemstudio.org/

- **results in partitioning of data space (voronoi cells)**

k-means Algorithm

Implement in homework 1

- **Steps**:
  1. **Initialization**: Randomly assign *k* data points as initial cluster centers (mean of the cluster)
  2. **Assignment step**: Assign each data point to the cluster with closest cluster center
  3. **Update step**: Calculate new centroids (mean points) of cluster
- **Iterate steps 2 /3** until convergence (no changes in centroid)



Convolution theorem

https://subscription.packtpub.com/book/big_data_and_business_intelligence/9781789343731/3/ch03lvl1sec25/convolution-theorem-and-frequency-domain-gaussian-blur

$$f(x,y) * h(x,y) \quad \Leftrightarrow \quad F(u,v)H(u,v)$$

Space convolution = frequency multiplication

The convolution theorem says that **convolution** in an image domain is equivalent to a simple multiplication in the frequency domain

Convolution

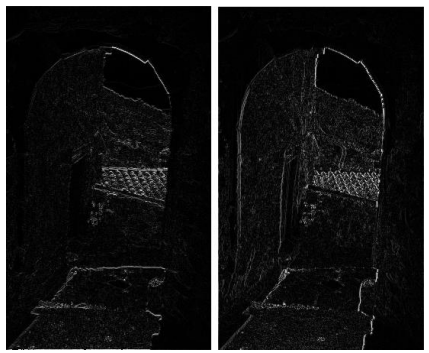
Convolution

- filter frequency characteristics of the image
  - general approach for image effects/analysis
  - convolution done by structural element (kernel)

Input image

Kernel

$$\begin{pmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{pmatrix}$$



Output image/
filtered image

- Compute new intensity value of central pixel as weighted sum of its neighbors

Convolution

- Convolution is multiplication of pixel and its neighbors by kernel matrix

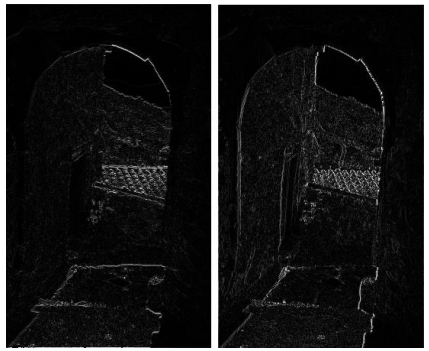
$$I' = \sum_{u,v} I(x-u, y-v) g(u,v)$$

I(x,y)

Input images

g(u,v,)

$$\begin{pmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{pmatrix}$$



Output image/
filtered image

What happens at borders?

- padding
- zero

Demo

http://setosa.io/ev/image-kernels/

Code yourself in first assignment

Image gradient - Sobel Filter

- Generation of an image gradient

$$\mathbf{G}_x = \mathbf{S}_x * A = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ 1 & 0 & -1 \end{bmatrix} * A$$

$$\mathbf{G}_y = \mathbf{S}_y * A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix} * A$$

Why is this an approximation of the gradient?

- Magnitude of gradient

$$\mathbf{G} = \sqrt{\mathbf{G}_x^2 + \mathbf{G}_y^2}$$

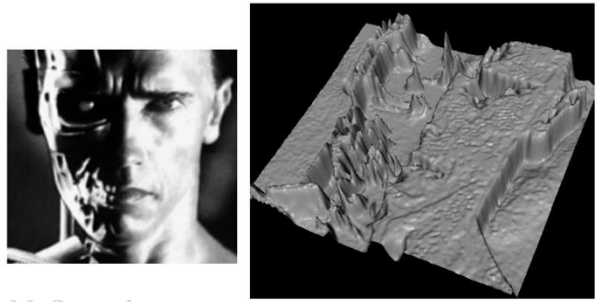
- Angle

$$\mathbf{\Theta} = \mathrm{atan2}(\mathbf{G}_y, \mathbf{G}_x)$$

- **Gradient intensities can be thresholded to detect edges**

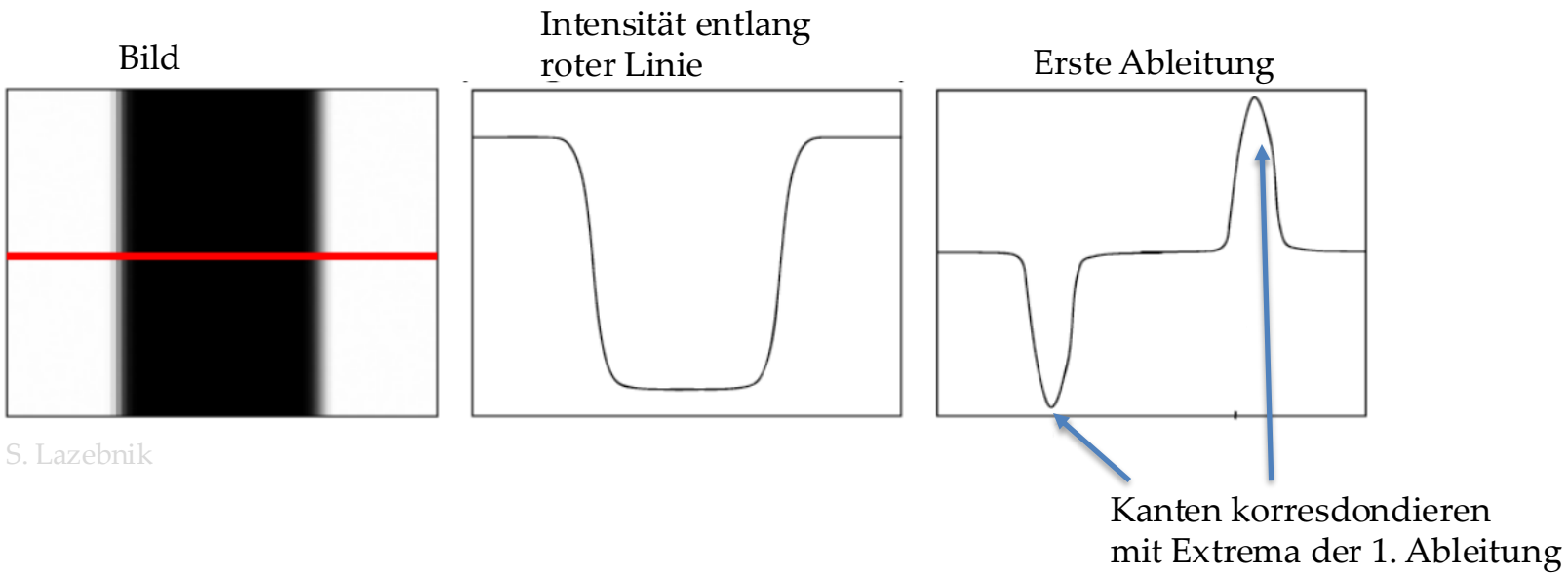
Bildgradienten

- Kanten sehen aus wie steile Abhänge



N. Snavely

- Orte plötzlicher Änderung der Bildintensität



S. Lazebnik

Bildgradienten

- Berechnung der diskreten Ableitung über finite Differenzen

$$\frac{\partial f(x,y)}{\partial x} \approx \frac{f[x+1,y] - f[x,y]}{1}$$

partielle Ableitung in x-Richtung

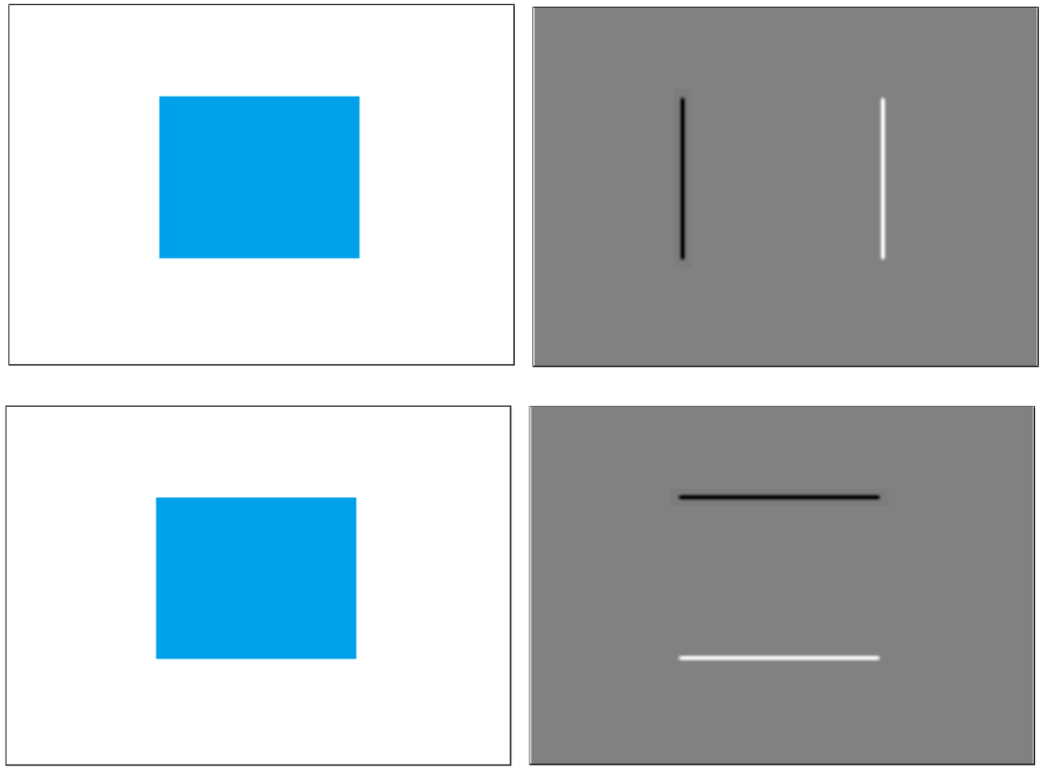
$$\frac{\partial f(x,y)}{\partial y} \approx \frac{f[x,y+1] - f[x,y]}{1}$$

partielle Ableitung in y-Richtung

- Wie könnte dafür ein Convolution-Filter aussehen?

Bildgradienten

- Convolution-Filter [-1,1] (horizontal), $[-1,1]^T$ (vertikal)



$$\frac{\partial f(x,y)}{\partial x} \approx \frac{f[x+1,y]-f[x,y]}{1}$$

$$\frac{\partial f(x,y)}{\partial y} \approx \frac{f[x,y+1]-f[x,y]}{1}$$

Fidler, Univ. Toronto

Finite Differenzen Filter

Prewitt: $M_x = \begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix}$; $M_y = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ -1 & -1 & -1 \end{bmatrix}$

Sobel: $M_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$; $M_y = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}$

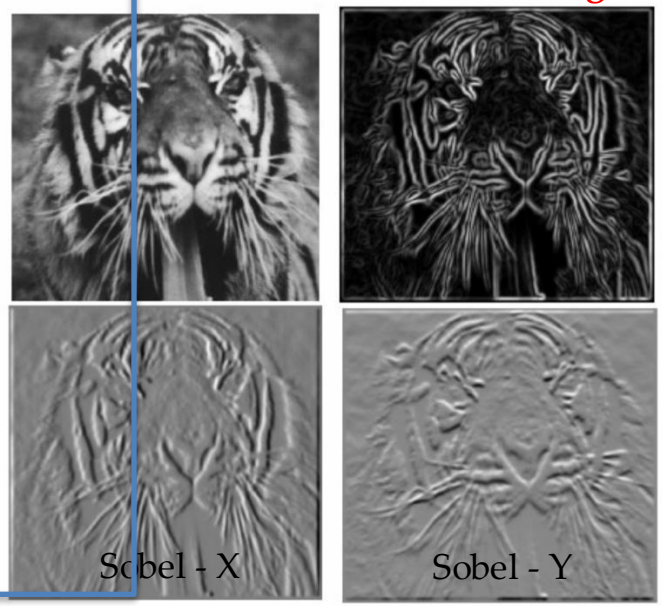
Roberts: $M_x = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$; $M_y = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

K. Grauman

Demo

Wichtige Änderungen im Bild!

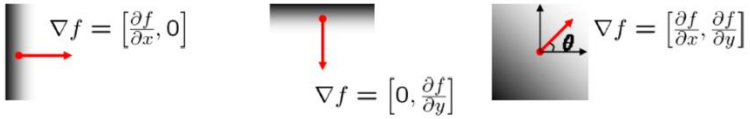
Gradientlänge



S. Lazebnik

Bildgradient als 2D Vektor:

$$\nabla f = [\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}]$$



Gradientenrichtung:
Richtung der Kantennormalen

$$\theta = tan^{-1}(\frac{\partial f}{\partial y} / \frac{\partial f}{\partial x})$$

Gradientenlänge: $||\nabla f|| = \sqrt{\frac{\partial f}{\partial x}^2 + \frac{\partial f}{\partial y}^2}$
Länge des Richtungsvektors

Separierbarkeit

- **Separierbarkeit** bezeichnet Eigenschaft, dass sich Impulsantwort eines zweidimensionalen Filters (z.B. mit Sobelkernel) durch Multiplikation zweier eindimensionaler Operatoren darstellen lässt
- Zweiter Operator wird auf Zwischenergebnis des ersten angewendet
- **ursprünglicher 2D-Filter in x- und y- Kern zerlegt und hintereinander auf Ursprungsbild angewandt**
- **Funktioniert nur, wenn Kernelmatrix Rang 1 hat und als Außerprodukt zweier Vektoren dargestellt werden kann**

$$K = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 \end{bmatrix}$$

- Warum?
  - Spart Rechenzeit, weil 2D Filter N^2 Multiplikationen + N^2 -1 Additionen benötigt
  - Separierte Variante nur 2N Multiplikationen und 2(N-1) Additionen
- Beispiel Sobel:

$$Sobel \quad M_x = \begin{pmatrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ 1 & 0 & -1 \end{pmatrix} \quad M_y = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{pmatrix} \quad M_x = \begin{pmatrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ 1 & 0 & -1 \end{pmatrix} * I = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} * \begin{pmatrix} 1 & 0 & -1 \end{pmatrix} * I$$

Canny-Edge Detection

Canny Edge

1. Apply Gaussian filter to smooth the image in order to remove the noise
2. Find the intensity gradients of the image

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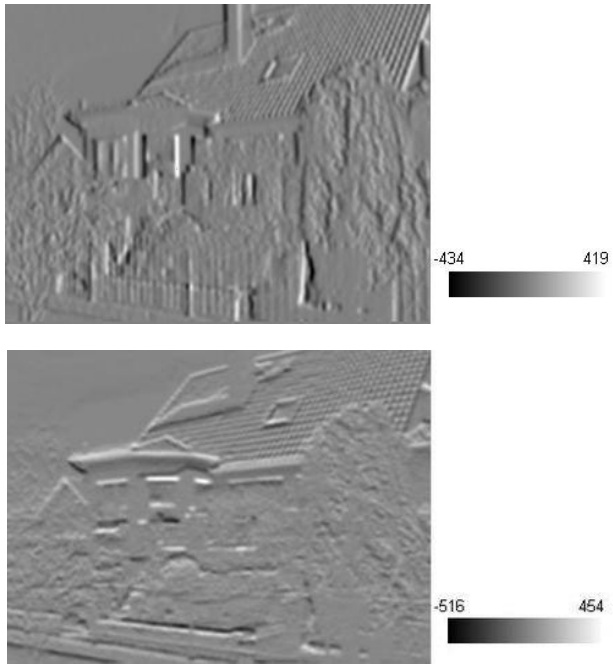

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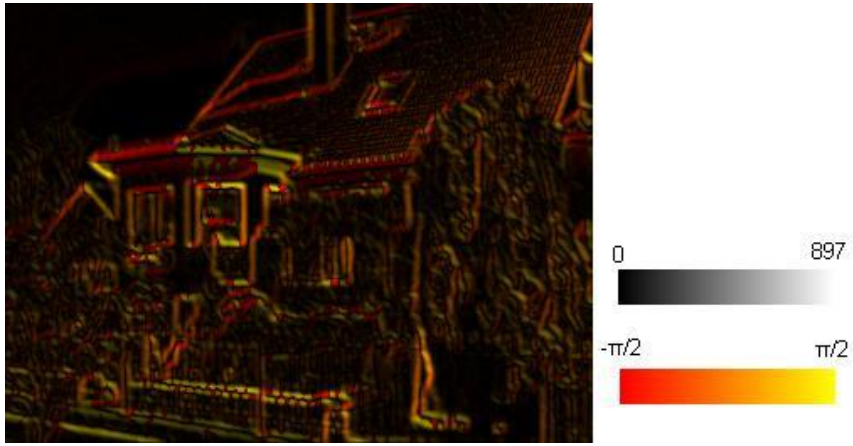

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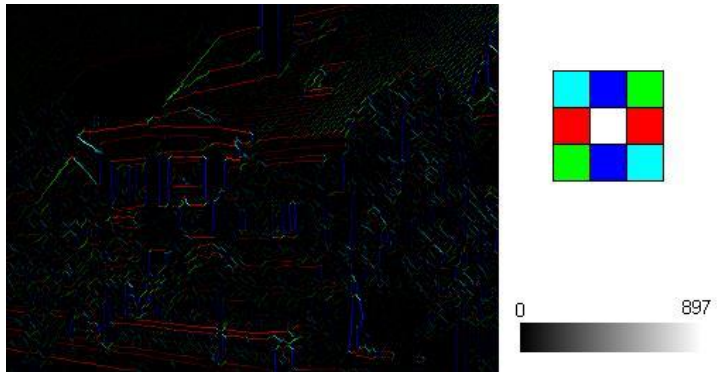
Canny Edge

3. Apply non-maximum suppression to get rid of spurious response to edge detection



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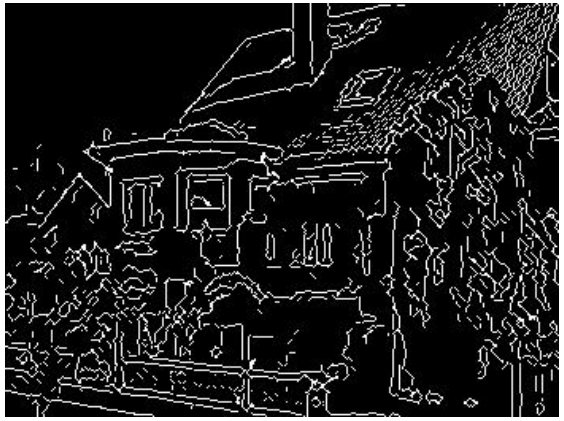
4. Track edge by hysteresis $T_1 < T_2$: Finalize the detection of edges by suppressing all the other edges that are weak and not connected to strong edges.



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